

Summary of sketching curve

discontinuous point $\rightarrow$ interception point (with x, y-axis) $\rightarrow$ Symmetry
 $\downarrow$
(concavity, inflection point) $f''(x) \leftarrow$ (extreme, monotonic) $f'(x) \leftarrow$ asymptotes

①. 确定 Domain $\triangle$ 任何和函数相关的题第一前都是写出定义域!

② Check whether the piecewise function is continuous and differentiable.

Step 1: continuous: $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$ $\triangle a$ is in domain of $f(x)$; $\lim_{x \rightarrow a} f(x)$ exists.

Step 2: differentiable: $\lim_{x \rightarrow a^+} f'(x) = \lim_{x \rightarrow a^-} f'(x)$ $\triangle \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$

$\triangle a$ 为函数分段的值.

Step 3: 补-同左 $x < a$ 和 $x > a$ 处 $f(x)$ continuous and differentiable.

③. Find asymptotes

Step 1: 先看 vertical asymptotes $\Rightarrow$ 出现在 domain 中断处及 domain 的 endpoint 处. (对于 ratio function, 分母为 0 对应
hole. 的 x 为 vertical asymptotes)
 $\lim_{x \rightarrow a^+} = \infty / -\infty$; $\lim_{x \rightarrow a^-} = \infty / -\infty$ $\triangle$ 只需满足其一即 $x=a$ 为 vertical asymptotes.

Step 2: 若 domain 的左右 endpoint 出现 $\pm\infty$ 时可能存在 horizontal asymptotes.

$$\lim_{x \rightarrow +\infty} f(x) = L \quad / \quad \lim_{x \rightarrow -\infty} f(x) = L.$$

Step 3: slant asymptotes: $\lim_{x \rightarrow \infty} f(x) = ax + b$ 求 a : $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = a$. 求 b : 代入 $a \Rightarrow \lim_{x \rightarrow \infty} f(x) - ax = b$

④. Intervals of Increase or Decrease:

Step 1: $\triangle$ 若没有检验 domain 分段处是否为 differentiable 则先进行检验, 再使用 1st derivative test.

① 说明 domain 上函数均 diff. and continuous.

step 2: 分段求 $f'(x)$ (若 $f(x)$ 为分段函数)

step 3: 列表 eg:

x	range 1	point	range 2
$f'(x)$			
$f(x)$			

⑤. local extreme point. $\Rightarrow$ 可以从④中的表格中得到. ④记得检验端点.

Δ 题型: global extreme point 是否存在 $\Rightarrow \lim_{x \rightarrow \infty} / \lim_{x \rightarrow -\infty}$, 看是否有 infinite limit.

(b). inflection point / concave upward or downward

$\Rightarrow$ 二阶导 (前提: 二阶导存在).

Δ inflection point: 二阶导 $= 0$ 且左右凹凸性不同.